

Optimizing MRI Protocols (SNR, CNR, Spatial Accuracy)

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Overview

- Parameters Affecting Image Quality
- Manipulating Image Contrast (Standard)
- Classifications of Pulse Sequences
- Relationships Between Resolution & SNR
- MR Image Speed
- Manipulating Image Contrast (Advanced)

Image Quality

- Image Contrast
- Image Resolution
- Signal-to-Noise Ratio
- Imaging Speed

Image Contrast

- Basic image contrast is effected by the amplitude and timing of the RF pulses used to excite the spin system.
- More advanced methods may use gradient pulses (to modulate motion) and alter tissue properties with exogenous contrast agents

Intrinsic (Tissue) Parameters

- Tissue Water Density (Proton Density)
- Longitudinal Relaxation Time (T1)
 - Magnetization transfer rate coefficients
- Transverse Relaxation Time (T2)
- Chemical Shift (water vs. fat)
- Tissue Motion
 - Macroscopic (flowing blood)
 - Microscopic (apparent diffusion coefficient)
- Tissue Susceptibility (air, fat, bone, blood)

Extrinsic (Selectable) Parameters

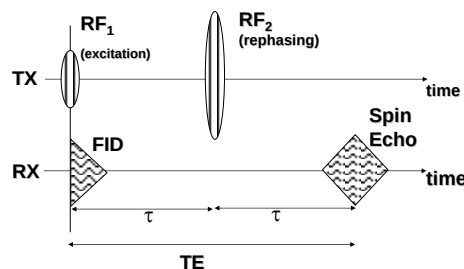
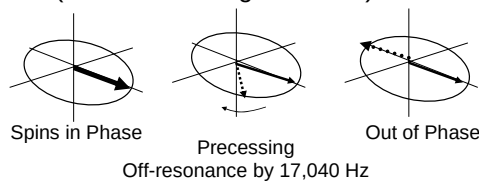
- Magnetic Field Strength
- RF Pulse timing
 - Excitation pulse repetition time (TR)
 - Time to echo (TE) & inter-echo spacing (T_{ie})
 - Inversion time (TI)
- RF pulse amplitude (flip angles)
- Gradient Amplitude & Timing
 - b-value
- RF pulse excitation frequency & bandwidth
- Receiver bandwidth

Spin (Proton) Density

Tissue	% Water	Tissue	% Water
Liver	70 %	Spleen	78%
Kidney	75%	Brain (GM)	85%
Muscle	73%	Brain (WM)	75%
Heart	78%	Tendon	60%

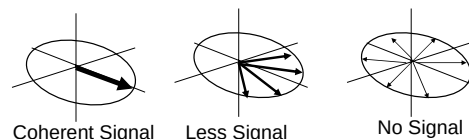
From Akber SF, *Physiol Chem Phys & Med* 1996; 28: 205-238.

Spin Echo Pulse Sequence

Out of Phase Spins
(Transverse Magnetization)

17, 040 Hz = 17, 040 cycles per second = $360 \times 17, 040$ degrees/sec

$$180^\circ \div (360^\circ \times 17, 040 \text{ degrees/sec}) = 3 \times 10^{-5} \text{ s} = 30\mu\text{s}$$

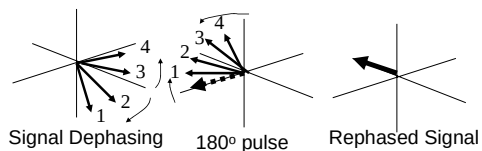
Dephasing of MR Signal
(Transverse Magnetization)

Signal decays exponentially:

$$(\text{after } 90^\circ \text{ rf pulse}) \quad M_{xy} = M_0 \exp [- TE / T_2^*]$$

where

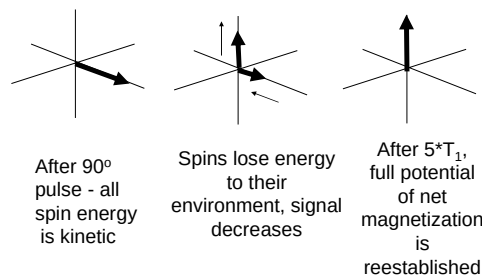
$$1/T_2^* = 1/T_2 + (\gamma/2\pi) * \Delta B_0 \quad \text{(Magnetic field inhomogeneity)}$$

Rephasing of the MR Signal
(Spin Echoes)

A rephasing 180° radio frequency pulse, changes the phase of the net magnetization vectors by 180° . Thus spins precessing in the same direction come together of moving apart. This forms a SPIN ECHO.

Longitudinal (T_1) Relaxation

Longitudinal relaxation is a **recovery** process:

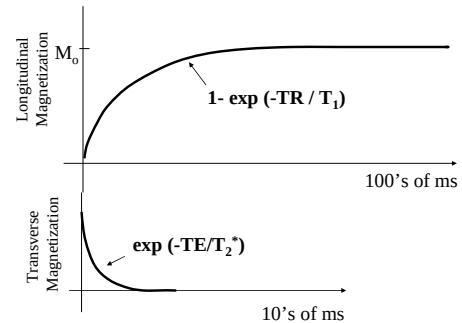


Comparison of T_1 , T_2 and for Various Tissues

Tissue	T_1 (ms)	T_2 (ms)
Liver	675 $\pm$ 142	54 $\pm$ 8
Kidney	559 $\pm$ 10	84 $\pm$ 8
Muscle	1123 $\pm$ 119	43 $\pm$ 4
Gray Matter	1136 $\pm$ 91	87 $\pm$ 15
White Matter	889 $\pm$ 30	86 $\pm$ 1.5

Akber, 1996 (at 63 MHz)

Time Scale of MR Relaxation Processes



Manipulating Contrast

- The “weighting” of image contrast
- Gradient Echo – manipulating image contrast by varying the flip angle
- Spin Echo – manipulating image contrast with 180° refocusing pulses
- Inversion Recovery – manipulating image contrast with 180° inversion pulses

Rules of Thumb

- TE controls T2 dependence
- TR controls T1 dependence
- T1 competes with T2 and proton density

“PD Weighted” = long TR, short TE

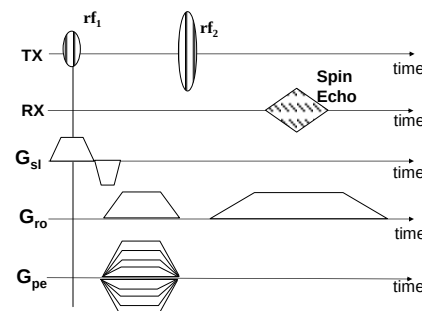
“T1 Weighted” = short TR, short TE

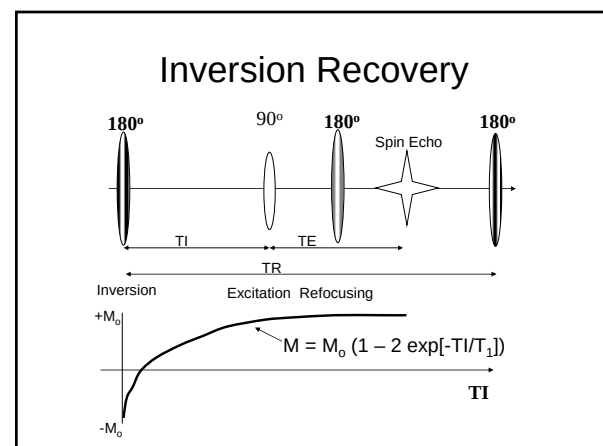
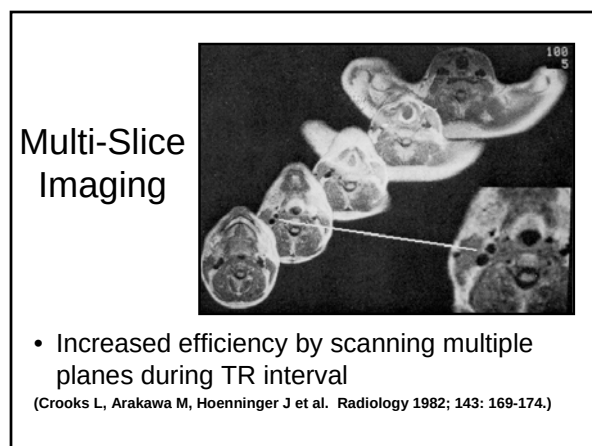
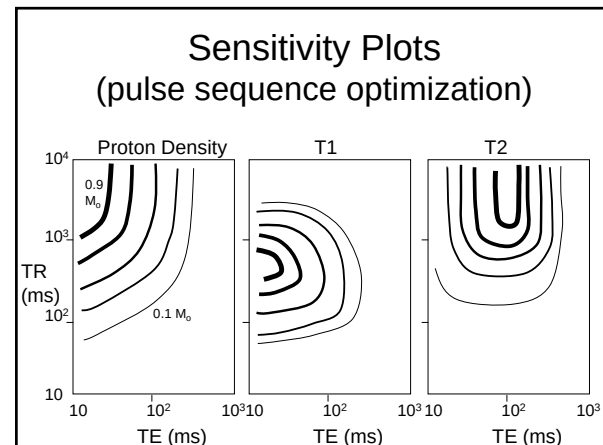
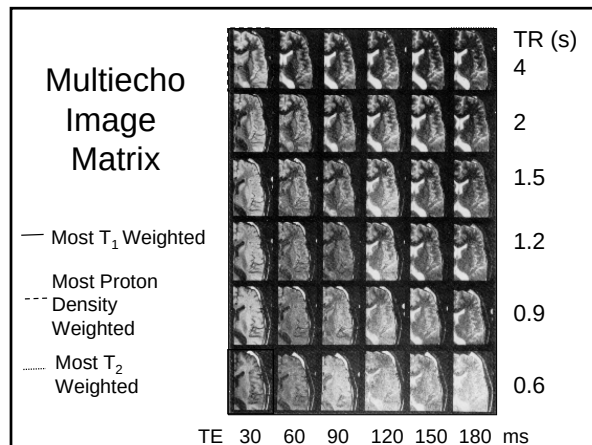
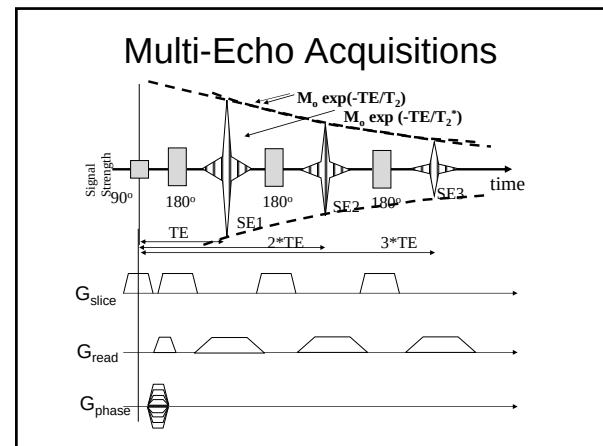
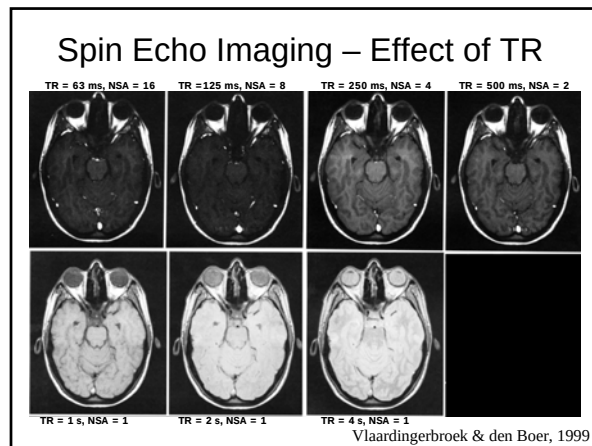
“T2 Weighted” = long TR, long TE

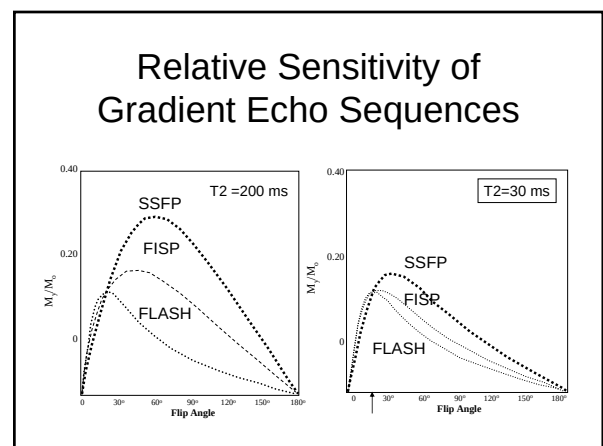
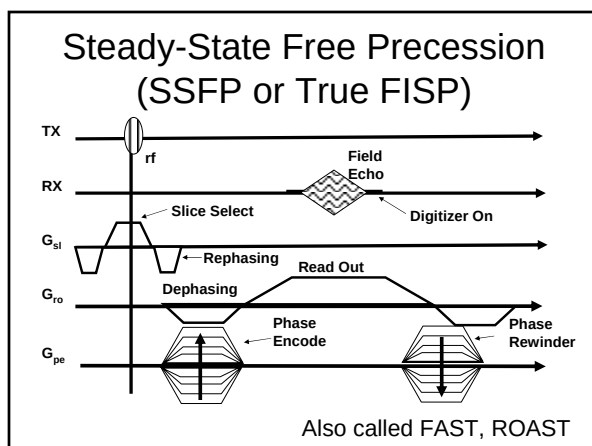
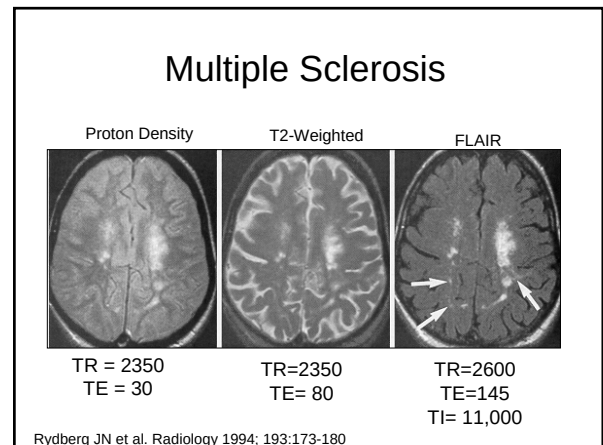
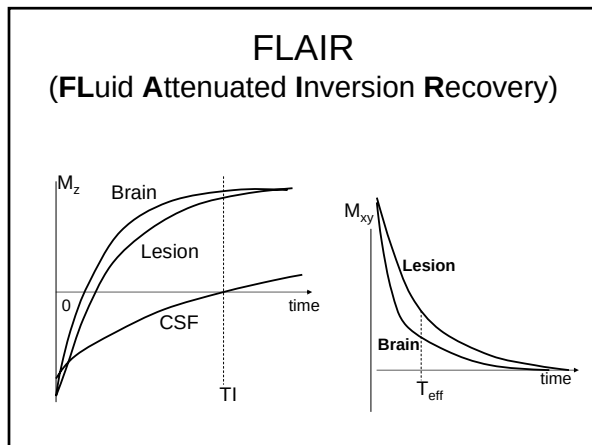
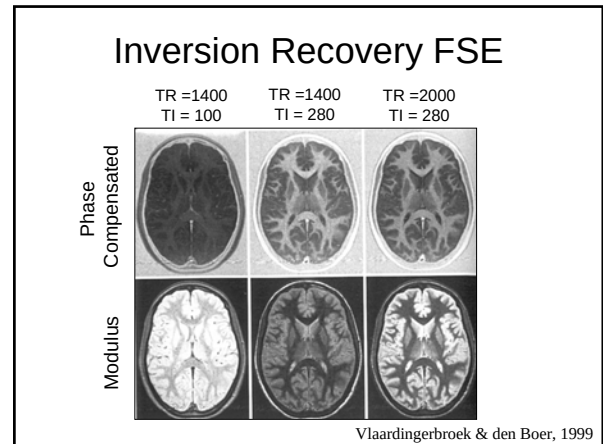
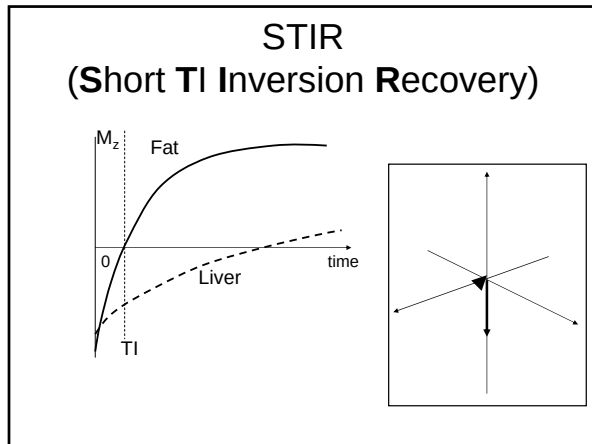
Pulse Sequence Classifications

Name	RF Pulses	Contrast Weighting	Application
Gradient Echo	One	T1 or T2*	Fast imaging (3DFT)
Spin Echo	Two or more	T1, PD or T2	conventional
Inversion Recovery	Three	T1 and T2	Excludes certain tissues

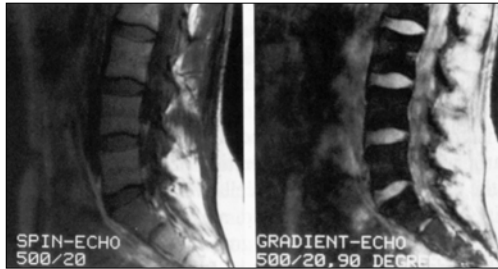
Spin-Echo Imaging Sequence







Gradient-Echo Imaging

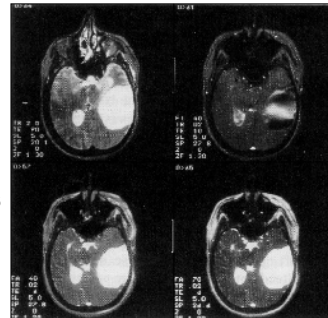


Reference: Wehrli, Fast-Scan Magnetic Resonance. Principles and Applications

Cystic Mass

Spin Echo
TR/TE = 2500/90
NSA = 1
5 mm

True FISP
TR/TE = 16/6
flip = 40°
NSA = 8
5 mm



FISP
TR/TE = 22/10
NSA = 8
8 mm

True FISP
TR/TE = 16/6
flip = 70°
NSA = 8
5 mm

Signal-to-Noise Ratio

- Signal-to-noise in an RF coil:

$$\frac{S}{N} \propto \frac{\omega_o M_o \int B_1(\vec{r}) dV}{\sqrt{4kT BW R_t}}$$

S = signal
N = noise
 ω_o = resonant frequency
 M_o = net magnetization
 $B_1(\vec{r})$ = RF field distribution
dV = elemental volume of tissue
k = Boltzman's constant
T = patient temperature
 BW = receiver bandwidth
 R_t = total resistance (coil + patient)

SNR and Imaging Parameters

$$SNR \propto \frac{FOV_{ro}}{\sqrt{M_{ro}}} \frac{FOV_{ph}}{\sqrt{M_{ph}}} \cdot \Delta z \cdot \frac{\sqrt{NSA}}{\sqrt{BW_{rx}}}$$

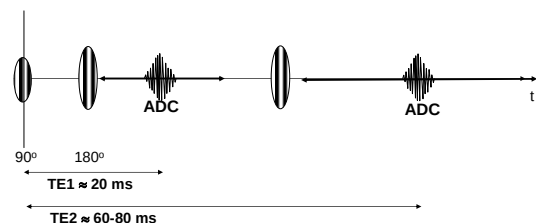
FOV = field of view M = matrix size n = number of signals averaged
 BW_{rx} = receiver bandwidth ro = read out (frequency encoding) direction
 Δz = slice thickness ph = phase encoding direction

Matched Bandwidth

- Bandwidth on ADC_1 is limited by finite T_{max} & desire to minimize TE_1 .
- T_1 weighting and SNR is sacrificed more by T_2 decay than by BW
- T_2 -weighted image - lots of time for ADC_2 , poorer SNR. It would be nice to decrease BW (increase T_{max})

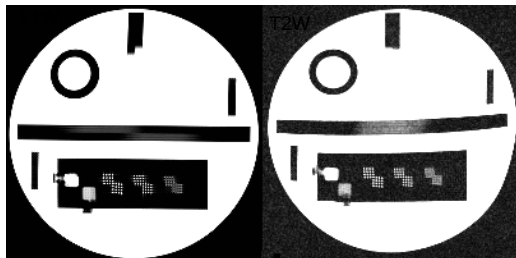
Mugler & Brookeman, Magn Reson Imag 1989; 7:487-493.

Matched Bandwidth



Mugler & Brookeman, Magn Reson Imag 1989; 7:487-493.

Matched Bandwidth = Unmatched Distortion



- Smaller BW on T2W image leads to increased image distortion

Chemical Shift

- Chemical shift (fat-water) ~3.5 ppm

– At 1.5T:

$$\Delta f_{\text{fat-water}} = 3.5 \times 10^{-6} \cdot \frac{42.6 \text{ MHz}}{T} \cdot 1.5T \cong 220 \text{ Hz}$$

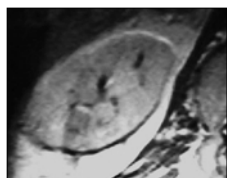
– At 0.5T:

$$\Delta f_{\text{fat-water}} = 3.5 \times 10^{-6} \cdot \frac{42.6 \text{ MHz}}{T} \cdot 0.5T \cong 73 \text{ Hz}$$

- ↓ field strength ↓ chemical shift

Chemical Shift Artifact

- Occurs in
 - Readout direction
 - Conventional SE
 - Phase encode direction
 - Echo-Planar
- Controlled by
 - Fat Pre Saturation
 - STIR sequence



Chemical shift artifact seen in axial image of kidney

Contrast-to-Noise Ratio

$$CNR = \sqrt{n} \frac{(I_a - I_b)}{\sigma}$$

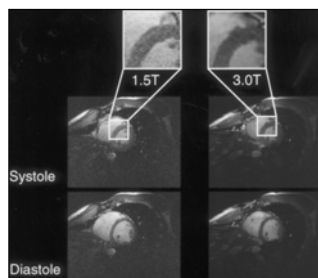
σ = standard deviation of the background signal (the noise)

n = number of signals acquired

I_a, I_b = signal intensities from tissues "a" and "b"

Steady state free precession (SSFP)

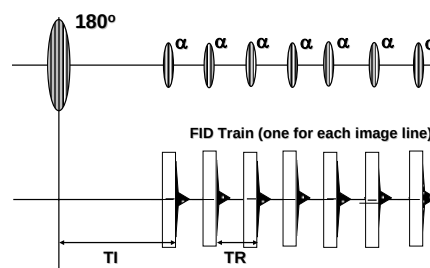
Signal is produced by rapidly applying balanced gradients and RF pulses. Use to maximize the overall signal.



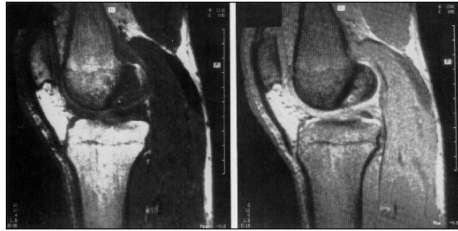
- FIESTA (fast imaging employing steady state acquisition)
- Balanced FFE (fast field echo)
- True FISP (fast imaging with steady precession)

Hinton et al. *Invest Radiol* 2003; 38:436

IR Turbo-FLASH



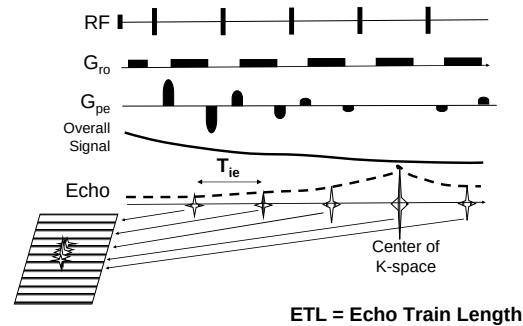
3D FLASH vs. 3D Turbo FLASH



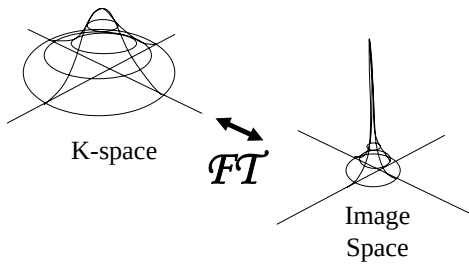
IR Turbo FLASH
 $T_I = 500$ ms, $TR/TE = 10/4.5$
 $NSA = 1$, flip = 10°

FLASH
 $TR/TE = 12/4.5$
 $NSA = 1$, flip = 8°

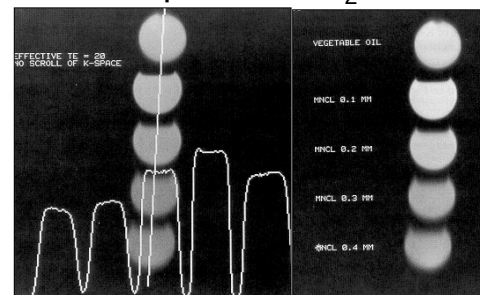
FSE Pulse Sequence



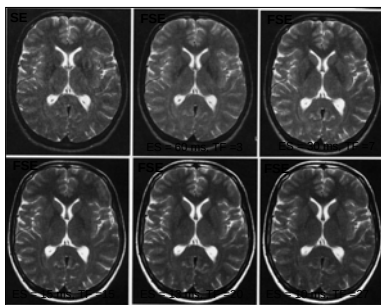
Things That have "Blurry" point spread functions are "Sharp" in the Image



FSE Point Spread Function Depends on T_2



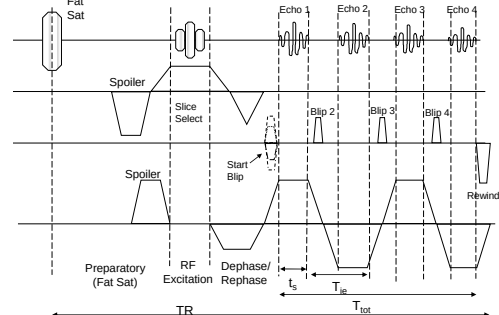
Effect of Echo Spacing



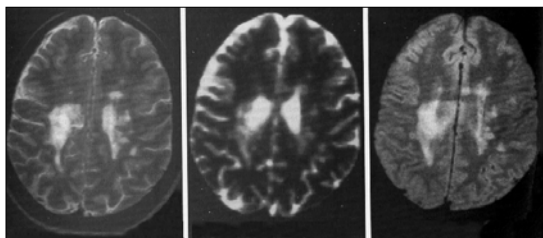
Signal-to-noise decreases for short T_2 tissues (gray & white matter) leading to a decrease in spatial resolution

Vladerenbroek & den Boer, 1999

Multi-Shot EPI Pulse Sequences

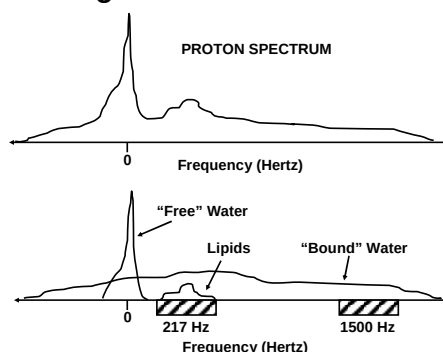


Comparison of EPI and SE

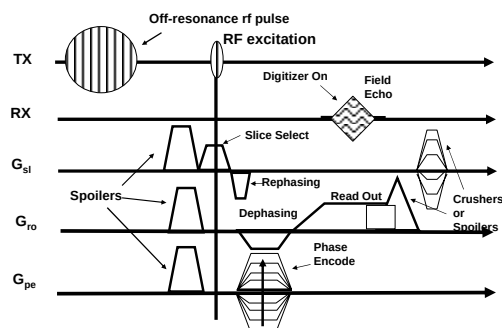


T2-weighted SE	Single shot EPI	Nine-shot EPI
TR 2400, TE 80	TE =80	TR 3000, TE 21
192x256; 7:44	128x128; 120 ms	252x256; 1:06

Magnetization Transfer



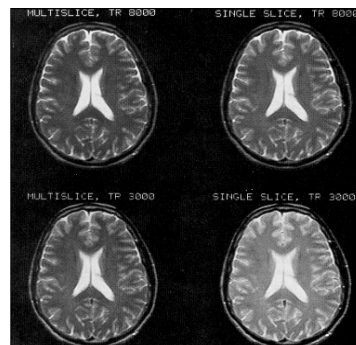
Gradient Echo with MTC Pulse



Magnetization Transfer Contrast

Multislice FSE:

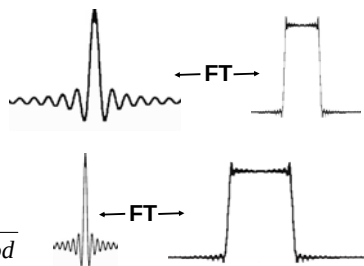
Magnetization Transfer Contrast Enhances T_2 Weighted Appearance



Slice thickness - RF bandwidth

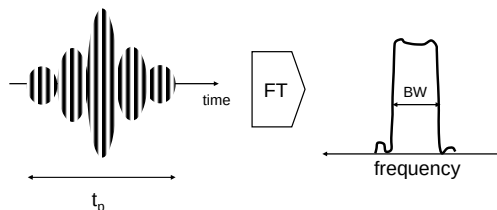
- thinner slices require longer RF pulses
- $\downarrow \Delta Z \downarrow \Delta \omega$

$$\text{Bandwidth} = \frac{1}{\text{period}}$$



Crafted RF Pulses

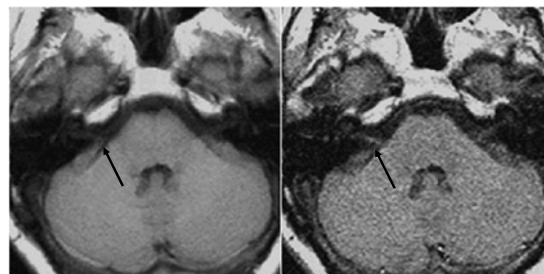
A sinc function ($\frac{\sin x}{x}$) envelope on the r.f. pulse produces a nearly square excitation profile of the phantom.....



MRI Slice Thickness

- Signal ramps have a slope of 10:1
- Signal from ramp is 10 x slice thickness
- Two ramps are used to compensate for in-plane rotation of the phantom
- Phantom does not compensate for tilting backwards or swaying left-right

Cranial Nerves



10 mm slice

3 mm slice

Frequency Encoding & FOV

- The field of view (FOV_x) is determined by the range of frequencies in the x axis which are sampled

$$FOV_x = \frac{Bandwidth}{\gamma \cdot G_x}$$

- The receiver bandwidth is not the same as the RF bandwidth
 - $\uparrow BW \uparrow FOV_x$
 - $\downarrow G_x \uparrow FOV_x$
- FOV must be sufficient to cover all tissue in the slice to avoid aliasing

Phase Encode Gradient

The maximum phase-encoding gradient is :

$$G_{ph-max} = \frac{2\pi}{\gamma} \cdot \frac{BW}{FOV} = \frac{\pi}{\gamma} \cdot \frac{N}{FOV \cdot T_y}$$

where FOV is the desired field of view

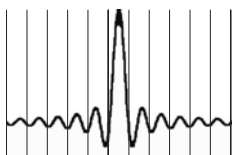
The phase-encoding gradient increment is :

$$\Delta G_{ph} = \frac{\pi}{\gamma} \cdot \frac{1}{FOV \cdot T_y}$$

- Note that a factor of two is gone from G_{ph-max} because G_{ph} runs from $-G_{ph-max}$ to G_{ph-max}

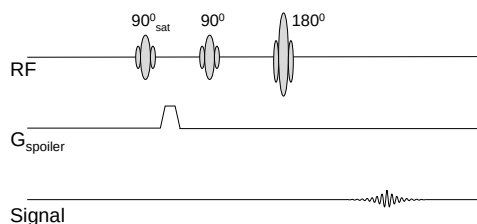
Sampling the Signal

- Each signal is digitally sampled to produce one line in the k -space matrix



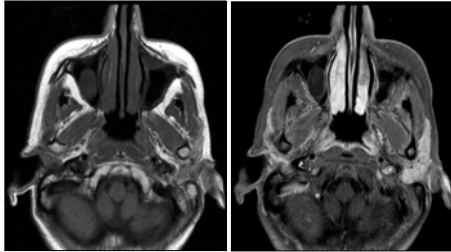
$$\begin{aligned} \Delta k_x &= \gamma \cdot G_x \Delta t_x \\ \Delta k_x &= \frac{MHz}{T} \cdot \frac{mT}{m} \cdot msec \\ &= \frac{cycles}{m} \\ \Delta k_x &= \frac{1}{FOV_x} \end{aligned}$$

CHESS



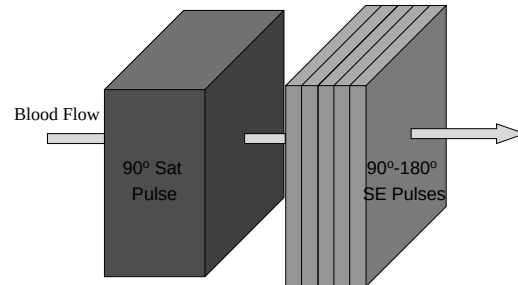
- This is typically accomplished by preceding a SE or FSE sequence with a 90° pulse that is frequency, not spatially, selective.

Fat Suppression

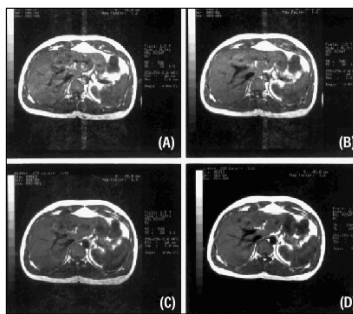


T1W images without (left) and with (right) fat suppression.

Spatial Saturation



Spatial Saturation



- No Sat Pulse
- Inferior Sat Pulse
- Superior Sat Pulse
- Inferior and Superior Sat Pulses

GE Signa Applications Guide, Vol 1

Gadolinium Contrast Agents

- GD-DTPA, Gd-DOTA, Gd-HP-DO3A:*

the old standards, generally diffusable but not in brain due to Blood Brain Barrier

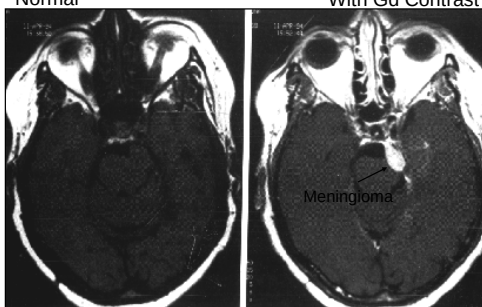
- In low doses decrease T1
- In higher doses decrease T1 and T_2^*

NEW INTRAVASCULAR AGENTS -

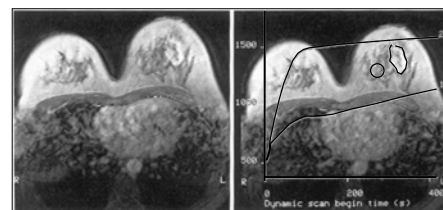
- Blood pool agents
 - Magnetic iron oxide particles
 - Bind serum albumin, have higher relaxivity

Gadolinium Contrast Agents

T₁ weighted Images



Dynamic Contrast Studies



- Dynamic study of breast tumor
- Curves denote difference between uptake of normal glandular tissue & lesion

Vlaardingerbroek & den Boer, 1999

Attenuation Due to Diffusion

$$A(TE) = A(0) \exp[-\gamma^2 G^2 D_{app} \delta^2 \alpha^2 (\Delta - \frac{\delta}{4})]$$

Where: $\alpha = \pi/2$;

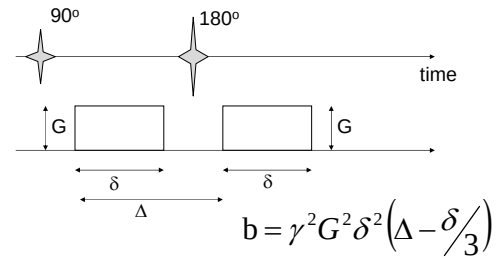
G is amplitude of diffusion sensitive gradient pulse;

δ is duration of diffusion sensitive gradient;

Δ is time between diffusion sensitive gradient pulses;

D_{app} is the apparent diffusion coefficient

Diffusion Weighted Imaging: Prototype Pulse Sequence

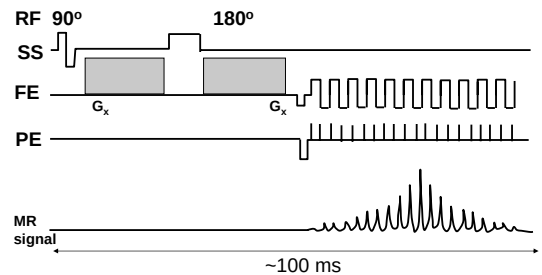


Stejskal EO & Tanner JE, 1965. 42: 288-292

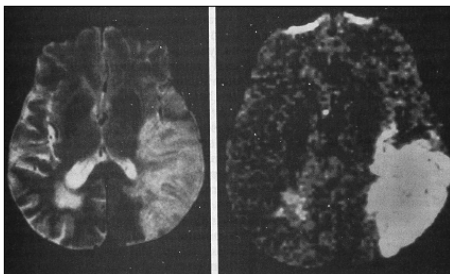
The b-value

- Controls amount of diffusion weighting in image
- The greater the b-value the greater the area under the diffusion-weighted gradient pulses
 - longer TE
 - stronger and faster ramping the gradients

Diffusion EPI Pulse Sequence



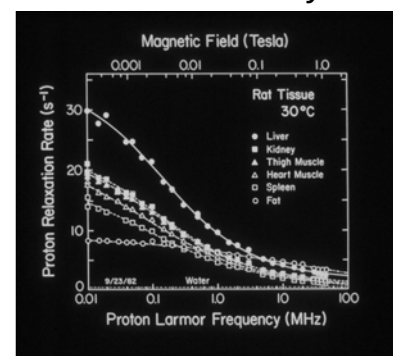
Diffusion Imaging of Infarcts



SE Echo Planar
TE = 80 ms

SE Echo Planar
b = 1205 s mm⁻¹

Biorelaxometry



Koenig, et. al, 1983

Specific Absorption Rate (SAR)

- The patient is in an RF magnetic field that causes spin excitation (the B1 field)
- The RF field can induce small currents in the electrically conductive patient which result in energy being absorbed.
- The RF power absorbed by the body is called the **specific absorption rate (SAR)**
- SAR has units of watts absorbed per kg of patient
- If the SAR exceeds the thermal regulation capacity the patient's body temperature will rise.

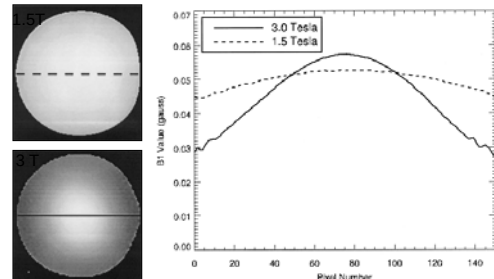
Scan Parameters Effecting SAR

- Patient size: SAR increases as the patient size increases – directly related to patient radius
- Resonant frequency: SAR increases with the square of the Larmor frequency (ω_0)
- RF pulse flip angle: SAR increases as the square of the flip angle (α)
- Number of RF pulses: SAR increases with the number of RF pulses in a given time

SAR Effects on Pulse Sequences

- Decrease number of slices per study
- Requires decreased flip angles, even in gradient echo sequences
- Forces increases in TR
- Limits use of Fast Spin Echo imaging

Uniformity at 3 Tesla



- B1 field maps in a conductive saline phantom (18 cm diameter)

RL Greenman et al. JMRI 2003, 17(6): 648-655

Summary

- RF pulses can be crafted to select slices
- RF pulse timing (TE, TR & TI) is used to manipulate soft tissue contrast in MR images
- Gradient echo imaging with partial flip angles is faster than spin echo imaging
- Spin Echoes correct for poor magnetic fields
- Inversion recovery can be used to get rid of fat or CSF from images
- Contrast agents are used to manipulate tissue T_1

Summary

- Presaturation schemes can effect flow, chemical shift and T1-weighted contrast
- Fast imaging methods are often applied to imaging physiology as well as anatomy
- Typically, SNR is reduced with increases in resolution, contrast and imaging speed
- Various image parameters will have to be adjusted for each magnetic field strength